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SCHOLARLY REVIEW

Quality Control of Protein Supplements: A Review

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The growing consumer awareness regarding health and fitness has been leading to a huge rise in the consumption of nutritional supplements and, consequently, to an increase in concerns about their quality. In this sense, one of the most consumed products is protein supplements and, despite being safer than other types of supplements, there are several studies showing incompatibilities between what is present on the labels and their actual content. Therefore, this review is focused on gathering information about the problems arising from poor manufacturing practices and inadequate quality control of sport protein supplements. These issues are mainly related to three aspects: reduction of the supplements' nutritional value, the presence of pharmacological substances, and contamination with microorganisms or toxic metals. Regarding the first aspect, reports about the "classic" addition of nitrogen-rich compounds to mask the protein content measured by the Kjeldahl method were discussed, as well as recent topics such as the addition of cheaper proteins to produce an "undetectable" adulteration in whey protein supplements. With respect to the presence of pharmacological compounds, it is a finding that is not very common in protein supplements; however, even trace amounts of foreign substances in this type of product may cause adverse effects to consumers, and, in the case of an elite athlete, may result in doping. Finally, we discuss about the contamination with microorganisms and toxic metals, this latter being a subject that should be further explored due to few studies in the literature.

***Keywords**:* adulteration, contamination, nutritional supplements, supplement quality control, whey protein

Nowadays, people are more often chasing the body considered as "ideal" by society. Likewise, the growing concern about image, esthetics, and healthy aging significantly increased the number of people who are engaged in exercise and, thus, increased the search for methods to improve performance and physical fitness (Sallis et al., 2016; Uhlmann et al., 2018). Consequently, the consumption of nutritional supplements has been continuously growing in recent years (Lukacs et al., 2018; Garthe, 2019). For instance, by 2018, the worldwide market was evaluated to be $115 billion (U.S. dollars) and is estimated to be worth twice this much in 2027 (Grand View Research, 2019, 2020). In this sense, one of the most used products is protein supplements (Knapik et al., 2016; Sánchez-Oliver et al., 2018).

The high boost in production, combined with poor manufacturing practices and the absence of specific regulations in many countries, has led to several reports in the literature about incompatibilities between what is on the label and the actual content of nutritional supplements (Fibigr et al., 2018; Marcus, 2016; Neves et al., 2015; Pawar & Grundel, 2017; Rocha & Amaral et al., 2016). Normally, these studies are more focused on herbals, prohormones, stimulants, and the presence of prohibited substances in sports (Ayotte et al., 2001; Backhouse et al., 2013; Cavalcanti et al., 2013; Geyer et al., 2008; Judkins & Prock, 2012; Krug et al., 2014; Martínez-Sanz et al., 2017; Mathews, 2018; Outram & Stewart, 2015). In fact, there is a great effort to monitor the misuse of supplements and drugs that contain (declared or not) substances that improve athletic performance, as each year, the list of banned substances and analytical methods are revised in a series of articles named Annual Banned-Substance Review—Analytical Approaches in Human Sports Drug Testing (Thevis et al., 2020). However, there are other types of occurrences that need attention. With this respect, although protein supplements are recognized to be safer compared with other types of supplements (Maughan, 2013), recent studies show that their quality control is an issue to be raised (De Ceglie et al., 2015; Garrido et al., 2016; Jiao et al., 2019; Roiffé et al., 2019; Sánchez-Oliver et al., 2018).

This article aimed to review the consumption of protein supplements, with a focus on sports, and also to discuss the issues related to poor manufacturing practices and inadequate quality control. In this respect, according to the reports in the literature, these issues were grouped as a reduction of the supplement nutritional value, the presence of pharmacological substances, and contamination with microorganisms or toxic metals.

Therefore, the discussions in this article are divided into four major sections. First, the composition of protein supplements and prevalence of use are presented. Second, the issue about protein supplement quality begins with reports in which there is a decrease of the supplement nutritional value, such as the addition of nitrogen-rich compounds to increase the total protein content assessed by the Kjeldahl method; the addition of cheaper protein sources (such as soy or rice) in whey protein (WP)-based supplements, and the loss of amino acid quality due to manufacturing and/or storage. Thereafter, the possible presence of pharmacological substances

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(as stimulants and prohormones) is presented and, finally, contamination with microorganisms and toxic metals is debated.

This is the first review relating exclusively to the quality control of protein supplements in the context of sports nutrition.

Literature Research

A review was carried out in the indexed bases ScienceDirect, PubMed, and Web of Science, using the terms "consumption of protein supplements," "protein supplement quality control," "protein supplement contamination," and "protein supplement adulteration." Moreover, references of pertinent articles were also reviewed. Only studies performed between January 2005 and July 2020 were evaluated. In this sense, 108 studies were selected for abstract reading, 81 were chosen for full-text evaluation, and 30 were included in this article. Only studies that conducted quality control analysis in protein supplements (i.e., evaluation of protein content, contamination with prohibited substances, toxic metals, and others) were considered fit for this purpose. It is worth mentioning that most of the excluded studies were related to the analysis of other types of supplements (e.g., creatine, herbals, and vitamins), or the type of supplement evaluated was not clearly specified.

Consumption of Protein Supplements

The consumption of dietary supplements is widely accepted and practiced by athletes, military personal, and fitness enthusiasts as a powerful mechanism to improve exercise performance (Maughan et al., 2018). Among the wide array of products available (herbals, multiminerals, multivitamins, prohormones, and various others), protein supplements are one of the most popular (Maughan, 2013; Schönfeldt et al., 2019).

Protein supplements are normally based on powdered caseins, egg, and soy proteins, as well as WPs, which are reconstituted with water or milk. Other types of supplements are also available, such as meat hydrolysates and pea and rice proteins (Corgneau et al., 2019; De Ceglie et al., 2015). Caseins are "slow-digesting" proteins rich in essential amino acids, which are released slowly after consumption; so, some athletes might ingest a casein supplement before bed to help in recovering and reducing muscle breakdown while sleeping (Joy et al., 2018). Soy proteins are a popular selling nonanimal protein source because of its high nutritional value, especially regarding the high content in arginine and glutamine, while egg proteins are a complete source of amino acids (Corgneau et al., 2019; De Ceglie et al., 2015; Wróblewska et al., 2018).

The most well-studied and commercialized protein supplement available on the market is WP (Borack et al., 2016; Chen et al., 2014), which is the soluble protein fraction in milk serum, obtained from the preparation of cheese and its derivatives, specifically during the casein precipitation step (Almeida et al., 2015; Haraguchi et al., 2006; Roiffé et al., 2019). Depending on the manufacturing process, WP may be classified as concentrate (30–85% protein), isolate (approximately 90% protein), or hydrolysate (smaller peptides that have faster intestinal absorption; Dalziel et al., 2016; Patel, 2015). Regardless of the form, WP has a series of highly digestible, rapidly absorbed, and high biological-value proteins (mainly α-lactalbumin and β-lactoglobulin) containing a large amount of essential amino acids (Garrido et al., 2016; Norton et al., 2016). These facts justify its high consumption, as several studies discuss that WP can increase muscle mass when combined with proper resistance exercise training and increase muscle and decrease fat mass when on a hypocaloric diet, in combination with exercise (Devries & Phillips, 2015; Marshall, 2004; Pasiakos et al., 2015).

The recommended daily dose of protein for adults is 0.8 g protein·kg^{-1}·day^{-1} (Phillips et al., 2016), while it has been suggested that athletes may have an increased dietary need of daily ingestion of 1.6 g protein·kg^{-1}·day^{-1} up to 2.2 g·kg^{-1}·day^{-1}, with no adverse effects (Maughan et al., 2018). In this respect, it is totally feasible for individuals who practice physical exercise to obtain their daily protein requirements through the consumption of whole foods. However, supplementation is much easier and faster than cooking, making it a practical way of accomplishing the recommended daily dose (Jäger et al., 2017). This practicality is one of the main factors that make protein supplements (and also other types of supplements) so widely consumed.

The consumption of nutritional supplements by professional athletes varies considerably depending on the type of sport, level of competition, and the definition of the supplements (Knapik et al., 2016; Garthe & Maughan, 2018). However, protein supplements are among the most prevalent in almost all studies. Knapik et al. recently conducted a systematic review and meta-analysis about the prevalence of dietary supplements and reported that 60% of athletes used this type of product, with multivitamins/multiminerals, protein, sport drinks, and sport bars being the most commonly consumed (Knapik et al., 2016). In a study of 778 Dutch elite and subelite athletes, Wardenaar et al. reported that 97.2% of the athletes had used nutritional supplements at some time during their sports career, whereas 84.7% indicated having used supplements during the last 4 weeks. In this context, the top five ranked sport nutrition products were isotonic sports drinks (44.1%), recovery drinks (33.5%), protein shakes (29.7%), energy bars (25.6%), and energy drinks (24.8%; Wardenaar et al., 2017). Heikkinen et al. evaluated the frequency of dietary supplement use among elite Finnish athletes in 2002 and 2009, reporting a high prevalence (81% in 2002 and 73% in 2009). In both years, the most popular products were protein supplements (47% in 2002 and 33% in 2009; Heikkinen et al., 2011). The use of protein supplements is also popular among young athletes, as Parnell et al. described a high rate of protein powder and bar consumption by Canadian young athletes (age between 11 and 25 years), while Whitehouse and Lawlis reported that 60% of a group of adolescent athletes (Australians aged 13–18 years) used protein supplements (Parnell et al., 2015; Whitehouse & Lawlis, 2017). Other reports about the consumption of supplements by athletes can be found elsewhere (Al Ghobain et al., 2016; Baltazar-Martins et al., 2019; Braun et al., 2009; Tawfik et al., 2016; Tsitsimpikou et al., 2009).

A high prevalence is also generally found in the military and gym users. For instance, in a study with 3,168 British Royal Army soldiers, a rate of supplement consumption of 38% was reported, reaching 54% when analyzing the last 12 months. The most used supplements were protein powders/bars (66%), isotonic sports drinks (49%), creatine (38%), recovery drinks (35%), multivitamins (31%), and Vitamin C (25%; Casey et al., 2014). Goston and Correia reported that gym users in a Brazilian population consumed five different products on a daily basis (which were classified as proteins and amino acids, isotonic drinks, carbohydrates, and natural/phytotherapeutic and multivitamin/mineral supplements), with protein supplements or amino acids being the most used products (58%; Goston & Correia, 2010). In a similar study, Sánchez-Oliver et al. evaluated the prevalence of protein supplement use by gyms users in Sevilla (Spain) and found that 28% of the 415 individuals consumed this type of product (being the most consumed supplement; Oliver et al., 2011).

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The high prevalence of the use of protein supplements across several exercising populations makes contamination/adulteration a concern. This process may affect their quality and safety, and the reviewed reports will be discussed in the next sections.

Decrease in Nutritional Value

Different parameters may be used to assess the nutritional quality of protein supplements, such as the protein content, amino acid profile, protein digestibility, bioavailability of essential amino acids, and others (Corgneau et al. 2019). Among these parameters, the actual protein content is one of the main quality indexes and an important message to communicate with consumers. Unfortunately, it is also one of the main problems reported in the literature regarding protein supplement quality control (Ingle et al., 2016; Jiao et al., 2019). Schönfeldt et al. reported that 68% of the 70 high-protein sports supplements evaluated presented actual protein content statistically different from the respective labels. In addition, they assessed the amino acid profile of 15 products, and 40% did not comply with the guidelines on the protein-scoring pattern for selected amino acids (Tryptophan, Threonine, Methionine + Cysteine, Valine, Phenylalanine + Tyrosine, Isoleucine, Leucine, Histidine, and Lysine; Schönfeldt et al., 2019). Almeida et al. evaluated the protein content of 10 WP samples bought in Brazil and 10 imported from the United States. They found that 11 of the 20 products (seven from Brazil and four from the United States) presented total protein content lower than the values described in the respective labels (Almeida et al., 2016a). In a similar study, Scarlato et al. analyzed 15 WP samples, with two presenting <75% of the declared protein content. In addition, 11 products had higher amounts of carbohydrates (two of them presented more than 250% of the labeled carbohydrate content; Scarlato et al., 2016).

However, even if a supplement has the measured protein quantity specified on the label in terms of nitrogen, it is possible the product does not contain what is expected. This is due to the fact that total protein content is usually measured by the Kjedahl method, which relies on total nitrogen determination as a marker to estimate protein quantity (Moore et al., 2010). This method does not distinguish protein nitrogen from nonprotein nitrogen sources; therefore, this drawback may be explored for potential adulteration of protein-based products. For instance, the producers may intentionally add nitrogen-rich compounds to mask the actual protein content (Garrido et al., 2016). A classic and tragic example of this type of adulteration was the melamine incident, which was made public in September 2008. In this incident, about 300,000 Chinese infants consumed milk-based products that had been deliberately adulterated with melamine, a multi-amine compound that is associated with kidney and urinary tract damage in humans. Six deaths were reported (Dorne et al., 2013; Gossner et al., 2009).

Regarding the addition of melamine in supplements, Gabriels et al. analyzed 138 supplement samples and found that 47% contained this compound. Although the levels detected were within the tolerable daily intake limit guidelines, melamine was not declared on the labels. They did not specify the number of protein supplements that were analyzed (Gabriels et al., 2015). There are other studies related to the analysis of melamine in supplements, and they do not report the presence of this substance (Beltrán-Martinavarro et al., 2012; Champagne & Emmel, 2011; Montesano et al., 2013). However, consumers should always be aware that it is possible to find melamine in protein-rich foodstuffs due to contamination (Lundy et al., 2011). Other nitrogen-rich compounds that have the potential to be used as adulterants to increase/correct protein content measured by the Kjeldahl method are cyanuric acid, creatine, glutamine, taurine, urea, and ammonium sulfate (Finete et al., 2013; Jiao et al., 2019). Cyanuric acid, for instance, may interact with melamine and form crystals in the urinary tract, leading to kidney problems (Reimschuessel et al., 2008). Therefore, an alternative to perform analyses that are more robust and less susceptible to fraud would be the combination of the classic Kjeldahl with other techniques, such as spectrometric or spectroscopic methods. In this sense, as it is not always possible to perform these types of analyses due to the costs (Chang & Zhang, 2017; Finete et al., 2013); however, some portable devices have been developed as inexpensive and practical alternatives (Karunathilaka et al., 2018; Silva & Rocha, 2020).

Another concern, especially when it comes to WP-based supplements, is the origin of the protein content. Garrido et al. used a proteomic approach to analyze 16 different brands of WPs and reported that six samples presented high concentrations of protein sources other than bovines, especially soybean, wheat, and rice, which are cheaper proteins with lower biological value. In addition, the same study discussed that some samples presented low levels of β-lactoglobulin, which might indicate the addition of whole milk instead of WP in these supplements (Garrido et al., 2016). In a related study, De Ciglie et al. assessed the composition of high-protein supplements commonly used by athletes and also found divergences. In one of the analyzed supplements, no egg components were detected, although egg white was declared as the second most abundant ingredient in the product. Moreover, two supplements did not present the expected amount of caseins (De Ceglie et al., 2015). As proteomics is a relatively new area, it is a field of study that may be explored much more in the future for the quality control assessment of protein supplements, especially in relation to WP analysis.

As mentioned before, protein digestibility may also be used to evaluate the quality of protein supplements. This parameter is important to estimate protein availability after digestion, which reflects on the efficiency of protein utilization on diet (Food and Agriculture Organization of the United Nations Expert Consultation, 2011). In this respect, Almeida et al. evaluated the in vitro protein digestibility of eight WP samples obtained in the United States and seven from Brazil (Almeida et al., 2015). They reported that the values obtained were in agreement with those in the literature, but there was a high variation, which might be explained by differences in the overall composition of such products, particularly with respect to the protein quality. In this sense, U.S. supplements exhibited better nutritional quality based on in vitro digestibility than the Brazilian supplements. In addition, the Protein Digestibility-Corrected Amino Acid Scores were calculated, and the values of both supplement groups were acceptable. However, when the Protein Digestibility-Corrected Amino Acid Scores calculation was based on the suggestion for adult athletes, both groups exhibited suboptimal score values for several essential amino acids. In another study, Corgneau et al. analyzed the protein digestibility of six commercial protein supplements (casein, milk protein concentrate, egg white, pea protein isolate, WP concentrate, and soy protein isolate). Casein and WP presented values in agreement with the Food and Agricultural Organization, whereas the other four exhibited lower quality regarding the Food and Agricultural Organization's expectation. This was due to low digestibility (for egg white and milk concentrate) and/or the lack of specific essential amino acids for milk concentrate and pea isolate (Corgneau et al., 2019).

Finally, a subject that is rarely discussed regarding protein supplements' quality control is the nutritional loss during their manufacturing and/or storage. For example, when WP supplements

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are produced, they undergo heating, drying, and ultrafiltration steps, which may lead to partial denaturation of the proteins, impacting their nutritional value. One of the reactions induced by these processes is the Maillard reaction, which takes place between reducing sugars and amino acids, peptides, or proteins (Bernard et al., 2011; Martinez-Alvarenga et al., 2014). In this sense, during WP manufacturing or storage, lactose may react with lysine, blocking the lysine amino group and forming the so-called blocking lysine, a product that may affect protein bioavailability (Tamanna and Mahmood, 2015). In this respect, Sánchez-Oliver analyzed 52 WP samples and reported that half of them contained over 6% of blocked lysine, and 9% presented values exceeding 20% (Sánchez-Oliver et al., 2018). This may mean that supplements with a high percentage of blocked lysine have less nutritional value, but more studies are needed to determine whether this is an appropriate parameter, and also which percentage would be adequate for the quality evaluation of sport protein supplements. In addition, another product of the Maillard reaction used to evaluate thermal and/or storage damage in foods is furosine (Mehta & Deeth, 2016). In this sense, Rufian-Henares et al. analyzed the furosine content of thirteen protein supplements and reported that the quantity ranged from 2.8 to 1125.7 mg/100 g protein (150 mg/100 g protein average; Rufian-Henares et al., 2007). Yet, they established that the value of 70 mg furosine/100 g protein content could be set up for controlling the quality of milk-based sport supplements (in their study, five products presented higher content). However, there are no other studies discussing this value or its use as a main reference to assess the quality of sport supplements. Regarding storage, Le et al. evaluated that this process may alter the chemical and physical aspects of milk protein concentrates. The increase in furosine levels, the decrease in solubility, and changes in color and aroma might be markers of long-time storage (Le et al., 2011). Table 1 presents a general overview of the studies related to a decrease in the nutritional value of protein supplements presented in the literature (chronological order).

Presence of Pharmacological Substances

As mentioned before, protein supplements are, generally, safer than other types of supplements regarding the contamination/adulteration with pharmacological activity substances. The "high-risk dietary supplements" usually include anabolic-androgenic steroids and stimulants, with rates of contamination between 12% and 58% (Martínez-Sanz et al., 2017; Mathews, 2018). The presence of undeclared substances is mainly caused due to inefficient quality control, and it can be originated through the absence of good manufacturing practices, as in cases of cross-contamination during supplement production or the intentional addition of pharmacological substances to enhance the desirable effects.

Regarding protein supplements, in a recent study, Tsarouhas et al. investigated the presence of banned doping substances in dietary supplements consumed by 170 athletes, reporting that 9% of the users were found to consume supplements contaminated with anabolic steroids, prohormones, selective androgen receptor modulators, and aromatase inhibitors. Protein/amino acids and creatine supplements (which corresponded to 15% of the samples) were found free from prohibited substances (Tsarouhas et al., 2018). However, other recent studies reported the presence of substances with pharmacological properties in the most consumed protein supplement. WP. Roiffé et al. analyzed 11 different brands of WP and found seven adulterated samples: one showed the presence of both conivaptan and politiazide (diuretic agents), two contained politiazide and benfluorex (a stimulant), and the others presented either politiazide or benfluorex (Roiffé et al., 2019). Moreover, in an evaluation performed by the Clean Label Project (2018), 55% of the 134 protein samples contained high amounts of bisphenol A, an endocrine disruptor normally present in plastic products and heat-sensitive papers (Rocha & da Costa et al., 2016). The same substance was also found in a WP sample analyzed in another recent study (Costa et al., 2020). In this study, 44 WP samples were analyzed, with caffeine being detected in 14 samples and bisphenol A in one of them. Caffeine was considered a contaminant because it was not declared on the product labels. With respect to bisphenol A, the authors believe that its presence is most likely to be derived from the contact of the protein powder with the plastic where the product was stored. An overview of the discussed studies and other reports related to this topic are presented, in chronological order, in Table 2.

The greater concern about the presence of pharmacological substances in protein supplements is that they are usually consumed in high doses (about 25–30 g/day, but there are people who consume more). Therefore, the analytical tests performed must be much more sensitive, as even trace amounts of foreign substances may cause adverse effects on consumers and/or result in doping, when it comes to athletes. This fact is well illustrated in the review carried out by Maughan (Maughan, 2013). In his example, if a protein powder contained 90 ng/g of a steroid, such as nandrolone, the result would be negative in a test with a limit of detection of 100 ng/g. However, a 25-g portion of this product would deliver a dose of 2.3 μg of the steroid, which might result in a positive doping test for nandrolone. In this sense, it is suggested the use of supplements that have a seal of approval that guarantees that the product has been tested for the presence of prohibited substances. It is also suggested not to use products that make a great appeal to their physiological effects, such as high muscle mass gain and weight loss (Geyer et al., 2011; Mathews, 2018).

Contamination With Microorganisms and Toxic Metals

Other important, but less common, reports in the literature regard the contamination of protein supplements with microorganisms and toxic metals. Regarding the first, in 2013, a wide-scale recall of products sold by a New Zealand enterprise (the world's largest milk products exporter) was announced after safety tests that indicated the probable presence of bacteria that may cause botulism. The suspicious products were 80% WP concentrates that had been sold to third parties that used it to produce nutritional supplements and infant formula. After a huge investigation, the government was able to establish that no products had been contaminated. However, the economic impact was severe, as many countries, including China, temporarily banned the import of certain New Zealand dairy products (Stojkov et al., 2018).

With respect to the presence of toxic metals, they might be found in supplements due to environment contamination, fertilizers, and inappropriate storage (Filipiak-Szok et al., 2015; Ziarati et al., 2018). The already mentioned Clean Label Project study reported that 70% of the analyzed protein products contained detectable levels of lead, 74% contained cadmium, and 85% contained arsenic. However, they did not specify whether the levels were above acceptable levels. Moreover, they concluded that egg protein sources tested cleaner, whereas plant-based proteins were the worst (Clean Label Project, 2018). Yet, two recent studies reported that they could detect traces of toxic metals in

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Table 1 Summary of the Studies Related to the Decrease in Nutritional Value of Protein Supplements

Author, year	Country	Aim	Sample	Result
Rufián-Henares et al., 2007	Spain	Assess nutritional quality of milk-based sport supplements by furosine content	13 WP samples	Furosine content ranged from 2.8 to 1125.7 mg/100 g protein (150 mg average); five products above reference value (70 mg/100 g protein)
Le et al., 2011	Australia	Evaluate chemical and physical changes in milk protein concentrate powder during storage	Method was not applied to commercial samples.	During storage, the levels of furosine increase, the solubility decreases, and the color of the product may change (get browner).
Champagne & Emmel et al., 2011	United States	Develop a rapid screening test for melamine and other adulterants in raw materials of dietary supplements	84 supplements (seven protein)	Protein supplements free from melamine and other adulterants
Montesano et al., 2013	Italy	Assessment of melamine in protein supplements by simple HPLC-DAD	Method was not applied to commercial samples.	The developed method was fully validated and presents a simple, fast, and selective HPLC-DAD method for the analysis of the protein samples adulterated with melamine.
Almeida et al., 2015	Brazil	Evaluate the in vitro digestibility of commercial WP supplements	Seven WP products from Brazil; eight WP products from the United States	United States whey supplements exhibited better nutritional quality based on in vitro digestibility than the Brazilian counterparts; both groups presented suboptimal PDCAAS values for several essential amino acids when considering suggestions for adult athletes.
De Ceglie et al., 2015	Italy	Quality control of high protein content sport supplement by MALDI-TOF	Four protein supplements	One sample presented complete absence of egg components, indicating fraud; two samples did not present the expected amount of caseins.
Gabriels et al., 2015	South Africa	Verify the possible presence of melamine in dietary supplements	138 supplements; 33 from South Africa and 105 imported into the country (did not specify how many protein products)	47% presented melamine: 27 South African products (82%) and 50 supplements imported into the country (58%) All of them within the tolerable daily intake
Scarlato et al., 2016	Brazil	Determination of protein and carbohydrate content in WP supplements	15 WP samples	Two samples with <75% of the protein content specified on label; 11 products with higher amounts of carbohydrates (two presented more than 250% of the specified amount)
Almeida et al., 2016[a]	Brazil	Evaluation of the protein and amino acid profiles of different WP supplements	10 WP products from Brazil 10 WP products from the United States	11 of the 20 products (seven from Brazil and four from the United States) presented lower total protein content; supplements from the United States showed better nutritional quality as evaluated through the contents of total protein, α-lactalbumin, β-lactoglobulin, and branched chain amino acids.
Almeida et al., 2016[b]	Brazil	Use Raman spectroscopy in quality control of dietary supplements	18 supplements (did not specify the number of protein supplements)	The use of Raman combined with chemometric tools is useful for screening of adulteration of dietary supplements.
Garrido et al., 2016	Brazil	Application of the proteomics in WP supplements quality control	16 different brands of WP supplements	Six samples with high concentrations of protein sources other than bovines, especially soybean, wheat, and rice; eight samples with low levels of β-lactoglobulin, indicating the addition of whole milk instead of WP
Andrade et al., 2018	Brazil	Investigate the potential of FTIR-ATR for qualitative and quantitative analysis of protein supplements	38 WP supplements	FTIR-ATR associated with chemometric tools can be used with accuracy in determinations of protein content of the evaluated supplements.
Sánchez-Oliver et al., 2018	Spain	Determination of the damage suffered by protein powder supplements during heat treatment	52 protein samples	26 supplements contained over 6% of blocked lysine (9% presented values exceeding 20%); the supplements with the highest concentrations of blocked lysine were whey hydrolysates and peptides (12%).

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Table 1 *(continued)*

Author, year	Country	Aim	Sample	Result
Corgneau et al., 2019	France	Evaluate the nutritional quality of commercial protein supplements	Six different types of commercial protein supplements	The protein content of all supplements met expectations; casein and WP presented digestibility values in agreement with the FAO, whereas the other four (egg white, milk concentrate, pea and soy isolate) exhibited a lower quality.
Jiao et al., 2019	China	Detection of adulterants in WP by Raman spectroscopy combined with multivariate analysis	Method was not applied to commercial samples.	Raman spectroscopy combined with multivariate analysis can quickly detect three adulterants (creatine, taurine, and glutamine) in WP powder.

Note. HPLC-DAD = High Performance Liquid Chromatography with Diode Array Detection; MALDI-TOF = Matrix Assisted Laser Desorption Ionization Time of Flight Mass Spectrometry; FTIR-ATR = Fourier-Transformed Infrared spectroscopy using Attenuated Total Reflectance; WP = whey protein; FAO = Food and Agriculture Organization of the United Nations Expert Consultation.

Table 2 Summary of the Studies Related to the Contamination/Adulteration of Protein Supplements With Pharmacological Substances

Author, year	Country	Aim	Sample	Result
Judkins et al., 2007	United States	Detection of steroids and stimulant as contaminants in dietary supplements	58 supplements (six protein)	Six supplements were contaminated with steroids and two with stimulants. None of them were protein supplements.
Martello et al., 2007	Italy	Detection of anabolic steroids and ephedrine in nutritional supplements	64 supplements (six protein)	17 supplements were contaminated: 12 contained illegal steroids, five contained ephedrine; one protein supplement was contaminated with illegal steroid: 4-Oestren-3,17-dione.
Stepan et al., 2008	Czech Republic	Determination of anabolic steroids and related compounds in nutritional supplements	48 supplements (29 protein)	Three samples were positive; one protein sample was contaminated with nandrolone, testosterone, and dehydroepiandrosterone.
Neves & Caldas, 2015	Brazil	Evaluate the profile of dietary supplements on the Brazilian clandestine market	2,898 different dietary samples (it does not specify how many protein products)	180 products were adulterated; one protein product contained sibutramine.
Stajić et al., 2017	Serbia	Determination of higenamine in dietary supplements	19 dietary supplements (one was protein)	Two samples contained higenamine (the protein supplement was free from this substance).
Clean Label Project, 2018	United States	Quality control of top-selling protein supplements	134 protein supplements	74 samples tested high levels of bisphenol A.
Müller et al., 2018	Brazil	Analysis of presence of synthetic drugs in dietary supplements	113 supplements (it does not specify how many protein supplements)	14 samples contained hydrochlorothiazide; one of them was a protein supplement.
Tsarouhas et al., 2018	Greece	Analysis of nutritional supplements consumed by 170 adolescent athletes	15% proteins/amino acids and creatine	9% of the athletes used nutritional supplements containing prohibited substances; protein/amino acids and creatine supplements free from banned doping substances.
Roiffé et al., 2019	Brazil	Analysis of diuretics and stimulants in WP	11 WP samples	Seven samples adulterated: one with conivaptan and politiazide; two contained politiazide and benfluorex; and the others presented either politiazide or benfluorex.
Costa et al., 2020	Brazil	Analysis of anabolic agents in WP supplements	44 WP samples	Samples free of anabolic agents; 14 samples contained caffeine (not declared on label); one sample contained bisphenol A.

Note. WP = whey protein.

protein supplements (such as mercury, cadmium, lead, and others), but they were all within the acceptable limit (de Aquino et al., 2017; Elgammal et al., 2019). In this sense, the analysis of toxic metals in protein supplements (especially WP) is also an area that should be further explored, since there are only a few reports in the literature, whereas several studies have conducted research on metal content in other types of supplements and in milk products (Castro-González et al., 2018; Filipiak-Szok et al., 2015; Mahmoudi

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Table 3 Summary of the Studies Related to the Contamination of Protein Supplements With Microorganisms or Toxic Metals

Author, year	Country	Aim	Sample	Result
Aljaloud et al., 2013	Saudi Arabia	Microbiological quality control of dietary supplements sold in South Arabia	80 different dietary supplements (three protein samples)	One WP sample contaminated with *Staphylococcus aureus* (4.36 Log CFU/ml).
de Aquino et al., 2017	Brazil	Assessment of mercury content in WP samples	19 different WP brands	All mercury concentrations found were lower than the limit recommended by Brazilian regulation.
Clean Label Project, 2018	United States	Quality control of top-selling protein supplements	134 protein supplements	70% contained lead; 74% contained cadmium; 85% contained arsenic.
Molina et al., 2018	Brazil	Microbiological quality control of energetic and protein supplements	Nine different brands of supplements: six protein products and three energetics.	Two albumin samples contaminated with *Serratia* spp (2.5×10^2 to 2.06×10^3 CFU/mL (min-max); One WP sample contaminated with *Citrobacter freundii* (2.2×10^2 to 1.33×10^3 CFU/mL (min-max))
Elgammal et al., 2019	Egypt	Determination of metal content in WP samples from markets in Giza, Egypt	26 WP samples	Hg was not detected in any sample; Mn, Fe, Cr, Na, and Cu were detected but not declared on labels; all levels of Ca, Na, Mg, Fe, Al, Mn, Ni, Cr, Sn, Cd, and Pb were within the limits established by the European Commission; levels of Cu and Zn were slightly higher in one sample each.

Note. CFU = colony-forming unit; WP = whey protein.

et al., 2017; Poniedziałek et al., 2018; Shahbazi et al., 2016; Tumir et al., 2010; Tunegova et al., 2016; Ziarati et al., 2018). A summary of the studies regarding the microbiological and toxic metal contamination of protein supplements is present in Table 3.

Conclusion

As seen from the supplied material, there are several reports of poor manufacturing practices that may lead to a decrease in the product's nutritional value and, also, to the presence of foreign substances. In this sense, some adulteration/contamination techniques are already well known, such as the addition of nitrogen-rich compounds to "trick" the total protein content evaluation by the Kjeldahl method, while other areas can be further explored in order to assess the quality of protein supplements, such as the study of products from the Maillard reaction to assess the loss of nutritional value resulting from production/storage; the study of proteomics to assess the protein composition; and, finally, the evaluation of the presence of toxic metals.

As for athletes, who are generally the biggest consumers of supplements, they need to be aware that they might be consuming products with a lower nutritional value than expected, which may affect their diet and performance. In addition, protein supplements are safer compared with other products available on market; however, if there is even a trace amount of prohibited substances, this may result in an adverse analytical finding in an antidoping test. Therefore, it is recommended for them to use supplements that have a quality seal of approval.

A possible limitation of our work is that a number of articles do not specify the type of the supplements that were analyzed, as they treat all samples as just "nutritional supplements." Therefore, some studies that presented irregularities in protein supplements may not have been included in this review, since it was not specified in the article whether they evaluated protein samples.

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